Return of the Ring

Whole-Number Certificates for Every Endpoint Total

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Abstract

The earlier paper *One Ring to Host It All* gives a test for when a single repeating cycle of underlying states can carry a given visible process for a given time. The test is a certificate: two short lists of whole numbers satisfying four families of linear inequalities. That paper left one implication open. If the inequalities can be satisfied by real numbers, can they be satisfied by whole numbers?

We prove that they can, for every finite map, every duration and all integer data. For natural-number data, which covers every case of the earlier paper, the implication is also proved in the Lean 4 proof assistant, and the formal proof is in the archive. The paper proves more. Set the fourth family aside and call the first three the core. In one connected part of the visible rule, take any one set of endpoint counts and add them up. The totals reached by whole-number solutions of the core are exactly the whole numbers between the smallest and the largest total reached by real solutions. So one lower limit, upper limit or exact value for that total can be met in whole numbers whenever it can be met at all, and whole-number limits on individual entries are kept. The fourth family is a lower limit of this kind, which settles the open implication. For endpoints so far out on a branch that the step after their window is still off the cycle, the largest real total is already a whole number.

Exact examples show where these statements stop: totals taken across two separate parts, weighted totals, totals of continuing counts along a tree, and two limits on two different totals can each fail. One consequence for the earlier paper is that existence of a host is decided by a direct check and one linear-programming feasibility test.

Keywords: functional graphs; integer certificates; integer rounding; network flows; circular words; aggregate constraints

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Overview

One Ring to Host It All [3], called *One Ring* below, asks when one repeating cycle of underlying states can display a given visible rule, for a given number of steps, from at least a given number of starting positions. Its answer is a test. Apart from one case that is checked directly, a suitable cycle exists exactly when two short lists of whole numbers satisfy four families of inequalities. For each visible state the first list counts the starting positions whose correct run *continues* into another starting position, and the second counts those at which such a run *ends*. The second kind are the *endpoints*; their total is the number of separate runs from which the cycle is assembled.

Inequalities of this kind are easy to satisfy with fractions and can be hard to satisfy with whole numbers. *One Ring* did not need to settle whether fractions are enough, because it tests the whole-number question by another route, and it recorded the matter as open. The difficulty is one family of the four. Whenever starts are required on a loop of the visible rule, it asks that the endpoint counts around that loop add up to at least one, and a total of one can be made from halves.

Two small cases show what is at stake. In both there are two visible states, one underlying copy of each, a duration of one step and no required starts, and we ask in addition that the two endpoint counts add up to at least one. First let the visible rule swap the two states, so that they form one loop. Half an endpoint at each state, with no continuing positions, satisfies the first three families, and the two halves add up to one. Here whole numbers do as well: one endpoint at the first state and none at the second. Now let the rule leave each state where it is, so that there are two separate loops. Half an endpoint at each state is again allowed and the total is again one, but the only whole-number choice is no endpoint at all. A total taken around one loop can be rounded; a total taken across two loops cannot. *One Ring* asks for its totals loop by loop, which is the first kind.

This paper proves that the first kind always rounds, and then goes further.

- **The open implication holds** (Theorem 1.1). For every finite visible rule, every duration and all whole-number data, if the four families have a real solution they have a whole-number solution.
- **Every endpoint total rounds** (Theorem 1.3 and Corollary 1.4). Set the fourth family aside and call the first three the *core*. Work in one connected part of the visible rule and choose any one set of endpoint counts: those on the loop, those of one branch, all of them, or any other selection. The totals that whole-number solutions of the core reach are exactly the whole numbers between the smallest and the largest total that real solutions reach. So one lower limit, one upper limit or one exact value for that total can be met in whole numbers whenever it can be met at all, and whole-number limits on individual entries of the two lists are respected throughout. The fourth family is a lower limit on the total around a loop, so the first statement is a special case of this one.
- **Deep endpoints lose nothing** (Theorem 1.6). Call an endpoint deep when it lies so far out on a branch that even the step after its promised window is still off the loop. For any set of deep endpoints the largest total reached by real solutions of the core is itself a whole number.
- **The statements are sharp in several directions** (Section 8). Exact small examples show that weighted totals, totals of continuing counts along a branch, mixtures of the two lists, and two limits on two different totals can fail to round.

An example from *One Ring* shows the second statement at work. Three visible states follow one another in a loop, $0 \to 1 \to 2 \to 0$, with duration one step, at least one start required at each state, and 3, 4 and 4 underlying copies. With the fourth family set aside, real solutions of the core reach endpoint totals from 0 up to $5\frac{1}{2}$. Whole-number solutions reach $0, 1, 2, 3, 4$ and 5, and no other total:

endpoint total	0	1	2	3	4	5
continuing counts g	$(1,1,1)$	$(0,1,1)$	$(0,0,1)$	$(0,0,0)$	$(0,0,0)$	$(0,0,0)$
endpoint counts h	$(0,0,0)$	$(1,0,0)$	$(1,1,0)$	$(1,1,1)$	$(1,1,2)$	$(1,2,2)$

The half is lost at the top, as it must be, and nothing is lost in between. In this example the fourth family asks for a total of at least one. Restoring it removes the first column and leaves the totals 1 to 5.

The proof translates a certificate into a flow in a small network, as *One Ring* does. A fractional solution of the kind that has to be rounded lies part of the way between two whole-number flows, and these differ by one closed walk in the network. Following the whole walk would overshoot the total. The proof instead follows the walk from a chosen starting point until the total has risen by exactly the amount needed, and then returns to the start along arcs that represent spare capacity. The return costs less than one extra turn around the loop, and a simple potential function shows where to start so that the walked part is cheap enough. For lower limits the walk stops the first time the total reaches the wanted level; for upper limits and exact values, where the walk may wind backwards, it stops the last time. Figure 2 shows the two rules.

One consequence returns to the earlier paper. Since real solutions already guarantee whole-number ones, existence of a suitable cycle can be decided by the direct check and a single linear-programming feasibility test (Corollary 1.2). The published algorithm, which solves at most one network-flow problem for each visible state, remains correct and returns a certificate; the new test is an alternative way to decide.

The precise statements are in Section 1. The network is described in Section 2, and the proof occupies Sections 3–5. Section 6 treats deep endpoints, Section 7 two further totals that round, and Section 8 the examples that fail and the questions still open. Section 9 relates the result to established work. Section 10 states exactly what has been checked and how, and the statement after it records how the mathematics was produced.

1. The question and the results

1.1 The certificate and the open implication

We recall the setting of *One Ring* [3]. Let Y be a nonempty finite set and $f : Y \to Y$ a map. The data are a horizon $T \in \mathbb{Z}_{\geq 0}$ and vectors $a, m \in \mathbb{Z}^Y$ of demands and inventories. Vector inequalities are coordinatewise. Define the push-forward matrix P and the matrix K_T by

$$(Pu)_y = \sum_{z : f(z) = y} u_z, \qquad K_T = \sum_{j=0}^{T} P^j.$$

Every weak component of the functional graph of f contains exactly one directed cycle. Write $\mathcal{C}$ for the set of these cycles and put $\eta_C = 1$ if $\sum_{y \in C} a_y > 0$ and $\eta_C = 0$ otherwise. The certificate system of *One Ring* asks for $g, h \geq 0$ indexed by Y with

$$g + h \geq a, \tag{1}$$

$$Pg \leq g + h, \tag{2}$$

$$g + K_T h \leq m, \tag{3}$$

$$\sum_{y \in C} h_y \geq \eta_C \qquad (C \in \mathcal{C}). \tag{4}$$

Here g_y counts the selected positions of label y whose run continues into another selected position, and h_y counts those at which a run ends; we call g the *continuing* variables and h the *endpoint* variables. Theorem 2.1 of *One Ring* states that, for $0 \leq a \leq m$ with $\sum_y m_y > 0$, a host exists if and only if the inventory is one constant exact cycle, which is checked directly, or (1)–(4) have a solution in nonnegative integers.

Let $\mathcal{R}(f, a, m, T)$ be the set of real solutions $g, h \geq 0$ of (1)–(4). Section 6.5 of *One Ring* records as open whether a real solution implies an integer one. Its Theorem 4.2 proves this when the quota (4) is replaced by $h_y \geq 1$ for one named cycle vertex y, and hence when every cycle is a fixed point. The general case is the following.

Theorem 1.1 (Aggregate implication). *For every finite map $f : Y \to Y$, every $T \in \mathbb{Z}_{\geq 0}$ and all $a, m \in \mathbb{Z}^Y$,*

$$\mathcal{R}(f, a, m, T) \neq \emptyset \implies \mathcal{R}(f, a, m, T) \cap \mathbb{Z}^{2Y} \neq \emptyset.$$

Corollary 1.2 (Decision by one linear program). *For data in the domain of* One Ring*, a host exists if and only if the positive support of m is exactly one directed cycle of f on which m is constant, or the linear system* (1)–(4) *has a real solution $g, h \geq 0$.*

The system has $2|Y|$ variables. Its coefficients are at most $T + 1$ and are computed without iterating T times, as in Section 5 of *One Ring*. The algorithm of that paper is unaffected: it remains correct, and it returns an integer certificate.

1.2 Endpoint totals in one component

Theorem 1.1 is a consequence of a statement about a single component, which is the main result. From here on, unless said otherwise, $f : V \to V$ is a map on a finite nonempty set whose functional graph is connected. Let C be its unique cycle, $c = |C|$, and write

$$\varphi = f^{T+1}.$$

The vertices outside C are the *tree vertices*. The data are $T \in \mathbb{Z}_{\geq 0}$ and arbitrary $a, m \in \mathbb{Z}^V$. A *core certificate* is a pair $g, h \in \mathbb{R}^V_{\geq 0}$ satisfying demand (1), matching (2) and capacity (3); there is no quota in this definition. We call $(g + K_T h)_v$ the *usage* at v.

A *box* is a set of pairs (g, h) given by integer lower bounds and integer or infinite upper bounds on the individual coordinates g_v and h_v. The trivial box imposes nothing. Fix a box and write $\mathcal{K}$ for the set of real core certificates in it. Since $K_T \geq I$, every core certificate has $0 \leq g_v, h_v \leq m_v$, so $\mathcal{K}$ is a polytope.

For a set $S \subseteq V$ write

$$q_S(h) = \sum_{w \in S} h_w.$$

An *integer interval* is a set $I = \{t \in \mathbb{R} : L \leq t \leq U\}$ with $L \in \mathbb{Z} \cup \{-\infty\}$, $U \in \mathbb{Z} \cup \{+\infty\}$ and $L \leq U$. It may be a lower limit, an upper limit, both, or a single value.

Theorem 1.3 (Endpoint totals). *Let f be connected, and let T, a, m and the box be as above. Let $S \subseteq V$ be any set and I any integer interval. If some $(g, h) \in \mathcal{K}$ has $q_S(h) \in I$, then some $(g, h) \in \mathcal{K} \cap \mathbb{Z}^{2V}$ has $q_S(h) \in I$.*

Corollary 1.4 (The whole range). *If $\mathcal{K} \neq \emptyset$, then*

$$\{q_S(h) : (g, h) \in \mathcal{K} \cap \mathbb{Z}^{2V}\} = \mathbb{Z} \cap \Big[\min_{\mathcal{K}} q_S,\ \max_{\mathcal{K}} q_S\Big].$$

In particular the largest integer value of q_S is $\lfloor \max_{\mathcal{K}} q_S \rfloor$, the smallest is $\lceil \min_{\mathcal{K}} q_S \rceil$, and every integer between them is attained.

Corollary 1.5 (Rounding inside a box). *Let $(g^*, h^*) \in \mathcal{K}$. There is $(g, h) \in \mathcal{K} \cap \mathbb{Z}^{2V}$ with*

$$\lfloor g^*_v \rfloor \leq g_v \leq \lceil g^*_v \rceil, \qquad \lfloor h^*_v \rfloor \leq h_v \leq \lceil h^*_v \rceil \qquad (v \in V),$$

and $\lfloor q_S(h^*) \rfloor \leq q_S(h) \leq \lceil q_S(h^*) \rceil$. If $q_S(h^*)$ is an integer, the total is kept exactly.

Corollary 1.5 strengthens the coordinatewise rounding of *One Ring* (its Corollary 3.3) by keeping one endpoint total as well. Taking $S = C$ in Theorem 1.3 gives every integer quota on the cycle endpoints, and $S = V$ gives the total number of endpoints. Sets that mix cycle vertices and tree vertices at any depth are included.

Theorem 1.3 adds one interval on one unweighted total to the core. It does not assert that an interval on a second set can be kept together with the quota (4). For $S = C$ the two conditions are a single interval, which is how Theorem 1.1 follows.

Call $w \in V$, and the variable h_w, *deep* if $\varphi(w) \notin C$, that is, if w is a tree vertex at distance more than $T + 1$ from the cycle. For deep endpoints no rounding is needed at the top.

Theorem 1.6 (Deep endpoints)**.** *Let f, T, a, m and the box be as in Theorem 1.3, and let $\omega \in \mathbb{R}^V_{\geq 0}$ vanish outside the deep vertices. If $\mathcal{K} \neq \emptyset$, the maximum of $\sum_v \omega_v h_v$ over $\mathcal{K}$ is attained at a point of $\mathcal{K} \cap \mathbb{Z}^{2V}$. In particular, for a set S of deep vertices,* $\max_{\mathcal{K}} q_S$ *is an integer.*

Theorem 1.6 has no counterpart for minima: a single deep endpoint can have real minimum $1/2$ (Section 8). Theorem 1.3 then gives the ceiling.

The hypotheses matter. Connectedness cannot be dropped. Arbitrary weights cannot in general replace the unit coefficients of the selected variables; Theorem 1.6 is an exception for maxima. Continuing variables cannot in general be included, and two intervals on two different sets cannot in general be imposed together. Section 8 gives an exact example for each, and states two questions that remain open.

2. The network

2.1 Flows and certificates

The network is that of Section 4 of *One Ring*, with the slack of each inequality written as a flow on its own arc. It has three vertices J_v, K_v, L_v for each $v \in V$ and five arcs for each v:

Arc	From, to	Lower bound	Its flow in terms of (g, h)
σ_v	$J_v \to K_v$	0	$g_v + h_v - (Pg)_v$, the slack of matching
β_v	$K_v \to L_v$	a_v	$g_v + h_v$
g_v	$L_v \to K_{f(v)}$	0 or more	the continuing variable
h_v	$L_v \to J_{\varphi(v)}$	0 or more	the endpoint variable
ψ_v	$J_v \to J_{f(v)}$	0	spare capacity

On each arc g_v and h_v the lower bound is the larger of 0 and the lower bound of the box, and the upper bound is that of the box when it is finite. No other arc has an upper bound. A *flow* is a real vector x on the arcs, within these bounds, such that the flow leaving each vertex minus the flow entering it equals

$$m_v - \sum_{u : f(u) = v} m_u \text{ at } J_v, \qquad 0 \text{ at } K_v \text{ and } L_v.$$

A *circulation* is a real vector on the arcs, of any signs, with all these balances zero. We write g and h also for the parts of a flow on the arcs of those names. Figure 1 shows a small case.

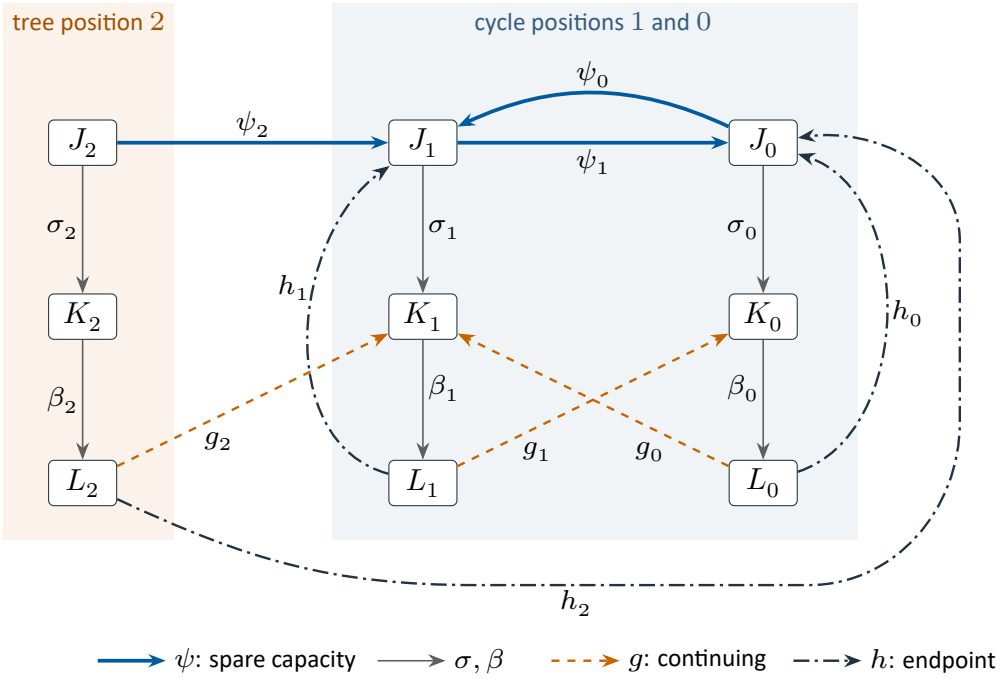


Figure 1: The network for a small map. Here $f(2) = 1$, $f(1) = 0$, $f(0) = 1$ and $T = 1$, so the cycle is $\{0, 1\}$, vertex 2 is a tree vertex, and $\varphi = f^2$ sends $2 \mapsto 0$, $1 \mapsto 1$, $0 \mapsto 0$. Each vertex v of the map has three network vertices J_v, K_v, L_v, all said to be at position v, and five arcs. The endpoint arc h_v jumps from L_v to the J vertex $T+1$ steps ahead; every other arc stays at its position or moves one step ahead. The advances are 1 on ψ_0, ψ_1, g_0, g_1 and on h_2, 2 on h_0 and h_1, and 0 on the other arcs.

Lemma 2.1 (Flows and certificates). *(a) For every flow x there is a number $\lambda(x)$, the* excess, *with*

$$x(\psi_v) + \big(g + K_T h\big)_v - m_v = \begin{cases} \lambda(x), & v \in C, \\ 0, & v \notin C. \end{cases}$$

The pair (g, h) of a flow is nonnegative, lies in the box, and satisfies demand, matching and the capacity inequalities at the tree vertices, and $(g + K_T h)_v \le m_v + \lambda(x)$ on the cycle. Conversely every pair $(g, h) \ge 0$ in the box and every number λ with these properties come from exactly one flow.

(b) The excess is an affine function of the flow and is an integer on integer flows. A flow with $\lambda \le 0$ gives a core certificate in the box, and every core certificate in the box is given by a flow with $\lambda = 0$.

(c) For a circulation D define $\lambda(D)$ by the same formula without m. Then $\lambda(x + D) = \lambda(x) + \lambda(D)$ for a flow x, and

$$c\,\lambda(D) = \sum_e \mathrm{adv}(e)\, D(e), \tag{5}$$

where the advance *of an arc is $\mathrm{adv}(\psi_v) = \mathrm{adv}(g_v) = 1$ for $v \in C$, $\mathrm{adv}(h_w) = \#\{t \in \{0, \dots, T\} : f^t(w) \in C\}$, and $\mathrm{adv}(e) = 0$ for every other arc. In particular $\lambda(D)$ is an integer when D is an integer circulation.*

Proof. Balance at L_v gives $x(\beta_v) = g_v + h_v$, and balance at K_v then gives $x(\sigma_v) = g_v + h_v - (Pg)_v$. So the bounds on β_v and σ_v are demand and matching. Write $u_v = (g + K_T h)_v$. Separating the term $t = 0$,

$$u_v - \sum_{f(z)=v} u_z = g_v + h_v - (Pg)_v - \sum_{\varphi(w)=v} h_w = x(\sigma_v) - \sum_{\varphi(w)=v} h_w.$$

Balance at J_v reads $x(\sigma_v) + x(\psi_v) - \sum_{f(z)=v} x(\psi_z) - \sum_{\varphi(w)=v} h_w = m_v - \sum_{f(z)=v} m_z$. Substituting, the quantity $\kappa_v = x(\psi_v) + u_v - m_v$ satisfies

$$\kappa_v = \sum_{z : f(z)=v} \kappa_z \qquad (v \in V).$$

A tree vertex with no preimage has $\kappa = 0$. The preimages of a tree vertex are tree vertices, so by induction along each tree towards the cycle $\kappa = 0$ at every tree vertex. On the cycle the equation then says that κ agrees at consecutive cycle vertices, so it is a constant $\lambda(x)$. The bound $x(\psi_v) \geq 0$ is capacity at a tree vertex and $u_v \leq m_v + \lambda$ at a cycle vertex. Conversely, from $(g, h) \geq 0$ and λ with these properties define the flow on β, σ, ψ by the displayed formulas; the three balances hold by the same computations. This proves (a), and (b) follows: $\lambda(x) = x(\psi_y) + u_y - m_y$ for a fixed $y \in C$ is affine with integer coefficients and integer constant.

For a circulation the same computation, with m absent, gives $\kappa_v(D) = D(\psi_v) + u_v(D)$, zero on tree vertices and equal to $\lambda(D)$ on the cycle; additivity is clear. Summing over $v \in C$,

$$c\,\lambda(D) = \sum_{v \in C} D(\psi_v) + \sum_{v \in C} D(g_v) + \sum_{w \in V} D(h_w)\,\#\{t \in \{0, \dots, T\} : f^t(w) \in C\},$$

which is (5). □

In the notation of *One Ring*, ψ, σ and $\beta - a$ are the slack variables on the base arcs, and λ is the multiple of the cycle direction that appears in the proof of its Lemma 3.2. That proof minimises λ over the flows. We do the same under one extra condition on the flow, which destroys the network structure and has to be handled by hand.

2.2 Positions

Say that J_v, K_v and L_v are *at position* v. A position is a cycle position or a tree position according as $v \in C$ or not. The *depth* of a position is its distance to the cycle. Every arc runs from a position v to the position $f^j(v)$ for some $j \geq 0$: $j = 0$ for σ and β, $j = 1$ for g and ψ, and $j = T + 1$ for h. Since f maps the cycle into itself, three facts follow.

Lemma 2.2 (Three facts about positions)**.** *(F1) An arc joining a vertex at a cycle position and a vertex at a tree position has its tail at the tree position. Its advance is nonnegative.*

(F2) An arc joining two vertices at tree positions has advance 0*.*

(F3) Let b be any position and y a cycle position, and let X be one of J_y, K_y, L_y. Follow the arcs $\psi_b, \psi_{f(b)}, \dots$ from J_b until J_y is first reached, then σ_y if $X \neq J_y$, then β_y if $X = L_y$. This forward path from J_b to X contains no g arc and no h arc, and its total advance is at most $c - 1$.

Proof. (F1) The head of an arc is at f^j of the position of its tail, and f^j maps cycle positions to cycle positions; all advances are nonnegative. (F2) Such an arc is not ψ_v or g_v with $v \in C$. If it is h_w, its head $\varphi(w) = f^{T+1}(w)$ is a tree position, so none of $f^0(w), \dots, f^T(w)$ lies on the cycle and $\mathrm{adv}(h_w) = 0$. (F3) The arcs ψ lead from J_b along the forward orbit of b, which enters the cycle and passes every cycle position. Until J_y is first reached the path uses at most $c - 1$ arcs ψ_v with $v \in C$, each of advance 1, and tree arcs ψ_v of advance 0. □

For a flow or circulation x write

$$q(x) = \sum_{w \in S} x(h_w)$$

for its *quota*; the set S is fixed in Sections 3 to 5.

3. One fractional edge and one rounding move

Let N be the set of all flows. It is a polyhedron, defined by the balance equations, the lower bounds of the table and the bounds of the box on the arcs g_v, h_v. All these data are integers.

A *signed simple circuit* Γ is a closed walk in the network, regarded as an undirected multigraph, that passes through no vertex twice. A loop is such a circuit, and so is a pair of parallel or opposite arcs. Each arc of Γ is traversed forwards or backwards; write Γ^+ and Γ^- for the two sets of arcs, and $\chi(\Gamma)$ for the vector that is $+1$ on Γ^+, -1 on Γ^- and 0 elsewhere. It is an integer circulation.

Lemma 3.1 (The flow polyhedron)**.** *(a) N contains no line, and every vertex of N is an integer point.*

(b) Every edge of N is a set $\{A_0 + s\,\chi(\Gamma) : 0 \le s \le \mu\}$, where A_0 is a vertex of N, Γ is a signed simple circuit, and μ is a positive integer or $\mu = \infty$ (with $s < \infty$).

Proof. These are standard facts about flows with integer data; we include the short argument. Every coordinate is bounded below, so N contains no line. At a point $x \in N$ call an arc *free* if x is not at a bound on it. The directions in which one can move from x within N, both ways, are the circulations supported on the free arcs; so the smallest face of N containing x has the dimension of that space of circulations.

(a) At a vertex no nonzero circulation is supported on the free arcs, so the free arcs contain no circuit and form a forest. The flow on every other arc is an integer bound. Removing leaves of the forest one at a time, each remaining value is determined by an integer balance, so it is an integer.

(b) On the relative interior of an edge the set of free arcs is constant and its space of circulations has dimension one. The support of a nonzero circulation contains the arcs of a signed simple circuit Γ, so that space is spanned by $\chi(\Gamma)$. Since N has no line, the edge has an end A_0, a vertex of N, and we may direct Γ so that the edge leaves A_0 along $+\chi(\Gamma)$. It stops, if at all, where some arc reaches a bound. As A_0 and the bounds are integers and the entries of $\chi(\Gamma)$ are $0, \pm 1$, that happens at an integer value of s. □

Lemma 3.2 (One edge)**.** *Let I be an integer interval and suppose $N_I = \{x \in N : q(x) \in I\}$ is not empty. Then λ attains its minimum over N_I at a point x^* that is either an integer point or of the form*

$$x^* = A + \frac{p}{k}\,\chi(\Gamma), \tag{6}$$

where A and $A + \chi(\Gamma)$ are integer points of N, Γ is a signed simple circuit, $k = q(\chi(\Gamma)) \ge 2$, and p is an integer with $0 < p < k$.

Proof. For $y \in C$, Lemma 2.1(a) gives $\lambda(x) = x(\psi_y) + (g + K_T h)_y - m_y \ge -m_y$ on N. So λ is bounded below on the nonempty polyhedron N_I, which contains no line, and therefore attains its minimum at a vertex x^* of N_I. If x^* is a vertex of N it is an integer point. Otherwise let F be the smallest face of N containing x^*; its dimension is at least one. The only conditions

of N_I beyond those of N are $q \geq L$ and $q \leq U$. If $q(x^*)$ lay strictly between L and U, a neighbourhood of x^* in F would lie in N_I. So $q(x^*)$ is one of the integers L, U; call it l. The points of F with $q = l$ lie in N_I and form a set of dimension at least $\dim F - 1$ through x^*. Since x^* is a vertex of N_I, the face F is an edge and q is not constant on it.

Write F as in Lemma 3.1(b) and $x^* = A_0 + s\chi(\Gamma_0)$. If s is an integer, x^* is an integer point. Otherwise, if $q(\chi(\Gamma_0)) > 0$ put $\Gamma = \Gamma_0$ and $A = A_0 + \lfloor s \rfloor \, \chi(\Gamma_0)$; if not, let Γ be Γ_0 reversed and put $A = A_0 + \lceil s \rceil \, \chi(\Gamma_0)$. In both cases $x^* = A + \theta\chi(\Gamma)$ with $0 < \theta < 1$, the points A and $A + \chi(\Gamma)$ are integer points of the edge, and $k = q(\chi(\Gamma)) > 0$. From $q(A) + \theta k = l$ we get $\theta = p/k$ with $p = l - q(A)$ an integer and $0 < p < k$. □

The reduction to a single edge after one extra linear condition is a known device; see Section 9. What is special to the present network is what can be done on that edge.

Lemma 3.3 (The rounding move)**.** *In the situation* (6) *put* $W = \lambda(\chi(\Gamma))$*, an integer. Let* D *be an integer circulation such that*

(1) $D(e) \geq -1$ *for* $e \in \Gamma^-$ *and* $D(e) \geq 0$ *for every other arc;*
(2) on every arc g_v *and* h_v, $D(e) = 0$ *or* $D(e) = \chi(\Gamma)(e)$*;*
(3) $q(D) = p$*;*
(4) $\lambda(D) \leq \lceil pW/k \rceil$.

Then $B = A + D$ *is an integer point of* N *with* $q(B) = q(x^*)$ *and* $\lambda(B) \leq \lceil \lambda(x^*) \rceil$.

Proof. B is an integer vector with the balances of a flow. On an arc g_v or h_v, $B(e)$ equals $A(e)$ or $(A + \chi(\Gamma))(e)$, and both respect the bounds. An arc e of the other three kinds has only a lower bound $\ell(e)$. If $e \notin \Gamma^-$ then $B(e) \geq A(e) \geq \ell(e)$; if $e \in \Gamma^-$ then $B(e) \geq A(e) - 1 = (A + \chi(\Gamma))(e) \geq \ell(e)$. So $B \in N$. Next, $q(B) = q(A) + p = q(x^*)$. Finally $\lambda(B) = \lambda(A) + \lambda(D) \leq \lambda(A) + \lceil pW/k \rceil = \lceil \lambda(A) + pW/k \rceil = \lceil \lambda(x^*) \rceil$, because $\lambda(A)$ is an integer. □

Taking all of Γ, that is $D = \chi(\Gamma)$, has quota k, which is too much for an upper limit or an exact value. Taking none has quota 0. The next section finds a circulation between the two.

4. The closing lemma

Lemma 4.1 (Closing lemma)**.** *Let* $S \subseteq V$*, and let* Γ *be a signed simple circuit with* $k = q(\chi(\Gamma)) \geq 2$*. Put* $W = \lambda(\chi(\Gamma))$*, an integer of any sign. For every integer* p *with* $0 < p < k$ *there is an integer circulation* D *with properties (1) to (4) of Lemma 3.3.*

Proof. Let Γ have M arcs $e_0, \dots, e_{M-1}$, in the order of traversal, with signs $\varepsilon_i = +1$ for a forward and -1 for a backward traversal, and let V_i be the vertex at which Γ stands before e_i. The V_i are distinct. Extend e_i, ε_i and V_i to all integers i with period M, and define Q and Θ on the integers by $Q(0) = \Theta(0) = 0$ and

$$Q(i+1) - Q(i) = \varepsilon_i \, [\, e_i = h_w \text{ with } w \in S \,], \qquad \Theta(i+1) - \Theta(i) = \varepsilon_i \, \mathrm{adv}(e_i).$$

Thus Q accumulates quota and Θ accumulates advance along Γ. By (5), $Q(i+M) = Q(i) + k$ and $\Theta(i+M) = \Theta(i) + cW$, so

$$\Phi(i) = cW\,Q(i) - k\,\Theta(i)$$

has period M. Two remarks will be used repeatedly.

(Q1) Q changes by at most one at each step.

(Q2) $Q(i+1) = Q(i) + 1$ only when e_i is a selected arc h_w traversed forwards, and then $V_{i+1} = J_{\varphi(w)}$. The quota rises only on arrival at a J vertex.

Let X be the set of indices i for which V_i is at a cycle position.

First case: X is not empty. Choose $i_0 \in X$ with $\Phi(i_0)$ least. Define the stopping index i_1 as follows (Figure 2).

- If $W \geq 0$, let i_1 be the least $i > i_0$ with $Q(i) = Q(i_0) + p$.
- If $W < 0$, let i_1 be the greatest $i < i_0 + M$ with $Q(i-1) = Q(i_0) + p - 1$ and $Q(i) = Q(i_0) + p$.

By (Q1), Q moves in steps of at most one from $Q(i_0)$ to $Q(i_0+M) = Q(i_0)+k > Q(i_0)+p$. So i_1 exists, $i_0 < i_1 < i_0 + M$, and the step into i_1 is upward; by (Q2), $V_{i_1} = J_b$ for some position b. Moreover

$$Q(s) \leq Q(i_0) + p - 1 \quad \text{for } i_0 \leq s < i_1, \qquad \text{if } W \geq 0; \tag{7}$$

$$Q(s) \geq Q(i_0) + p \quad \text{for } i_1 \leq s \leq i_0 + M, \qquad \text{if } W < 0. \tag{8}$$

The first holds because i_1 is the first index at which the level is reached. The second holds because after its last upward crossing of the level Q cannot be below it: to end at $Q(i_0) + k$ it would have to cross upward again.

Claim: $\Phi(i_1) \geq \Phi(i_0)$. If $i_1 \in X$ this is the choice of i_0. Suppose $i_1 \notin X$, so that b is a tree position.

If $W \geq 0$, let r be the greatest element of X below i_1; then $i_0 \leq r < i_1$ and V_s is at a tree position for $r < s \leq i_1$. The arc e_r joins V_r, at a cycle position, to V_{r+1}, at a tree position. By (F1) its tail is V_{r+1}, so it is traversed backwards and contributes $-\operatorname{adv}(e_r) \leq 0$ to Θ. Each arc e_s with $r < s < i_1$ joins two tree positions and has advance 0 by (F2). Hence $\Theta(i_1) - \Theta(r) \leq 0$, while $Q(i_1) - Q(r) \geq 1$ by (7). Therefore

$$\Phi(i_1) - \Phi(r) = cW\big(Q(i_1) - Q(r)\big) - k\big(\Theta(i_1) - \Theta(r)\big) \geq 0,$$

and $\Phi(i_1) \geq \Phi(r) \geq \Phi(i_0)$ since $r \in X$.

If $W < 0$, let r' be the least element of X above i_1; then $i_1 < r' \leq i_0 + M$, because $i_0 + M \in X$, and V_s is at a tree position for $i_1 \leq s < r'$. Each arc e_s with $i_1 \leq s < r' - 1$ joins two tree positions and has advance 0. The arc $e_{r'-1}$ joins $V_{r'-1}$, at a tree position, to $V_{r'}$, at a cycle position; by (F1) it is traversed forwards and contributes $\operatorname{adv}(e_{r'-1}) \geq 0$. Hence $\Theta(r') - \Theta(i_1) \geq 0$, and $Q(r') - Q(i_1) \geq 0$ by (8). As $W < 0$,

$$\Phi(r') - \Phi(i_1) = cW\big(Q(r') - Q(i_1)\big) - k\big(\Theta(r') - \Theta(i_1)\big) \leq 0,$$

and $\Phi(i_1) \geq \Phi(r') \geq \Phi(i_0)$ since $r' \in X$. The claim is proved.

Put $\alpha = \Theta(i_1) - \Theta(i_0)$. Since $Q(i_1) - Q(i_0) = p$, the claim reads $cWp - k\alpha \geq 0$, that is,

$$\alpha \leq \frac{p\,c\,W}{k}. \tag{9}$$

Now build D. Let ρ be the run $e_{i_0}, e_{i_0+1}, \ldots, e_{i_1-1}$ of Γ. It has fewer than M arcs, so it uses each arc of Γ at most once, with the sign it has in Γ, and it leads from V_{i_0} to J_b. The vertex V_{i_0} is at a cycle position; let Π be the forward path of (F3) from J_b to V_{i_0}. Let D be the sum of the signed incidence vectors of ρ and Π. It is an integer circulation.

(1) ρ contributes -1 only on arcs of Γ^-, once each, and Π contributes only nonnegative amounts.

(2) Π has no g arc and no h arc. On those arcs D is the contribution of ρ, which is 0 or the entry of $\chi(\Gamma)$.

(3) $q(D) = Q(i_1) - Q(i_0) = p$.

(4) By (5) and (F3), $c\,\lambda(D) = \alpha + \mathrm{adv}(\Pi)$ with $0 \le \mathrm{adv}(\Pi) \le c - 1$. Since $\lambda(D)$ is an integer, $c\,\lambda(D)$ is the one multiple of c among $\alpha, \alpha+1, \dots, \alpha+c-1$, so $\lambda(D) = \lceil \alpha/c \rceil$, and $\lambda(D) \le \lceil pW/k \rceil$ by (9).

Second case: X is empty. Then every arc of Γ joins two tree positions and has advance 0 by (F2), so $W = 0$. Let y be a position of least depth among the positions of the vertices of Γ. Every vertex of Γ is at a position u with $f^d(u) = y$ for some $d \ge 0$. Indeed, this holds at a vertex at position y, and it passes from each vertex of Γ to the next: if V_i is at u with $f^d(u) = y$ and V_{i+1} is at u', then either $u = f^j(u')$, so $f^{j+d}(u') = y$; or $u' = f^j(u)$ with $j \le d$, so $f^{d-j}(u') = y$; or $u' = f^j(u)$ with $j > d$, so $u' = f^{j-d}(y)$ would be a position of Γ nearer the cycle than y, which is excluded.

Choose i_0 with V_{i_0} at position y, and let i_1 be the least $i > i_0$ with $Q(i) = Q(i_0) + p$. As before $i_0 < i_1 < i_0 + M$ and $V_{i_1} = J_b$, where now $f^d(b) = y$ for some d. The arcs $\psi_b, \psi_{f(b)}, \dots$ lead from J_b to J_y through tree positions; add σ_y, and β_y, as needed to reach V_{i_0}. This path Π has no g or h arc and advance 0. With ρ the run from i_0 to i_1 and D defined as before, properties (1) to (3) hold for the same reasons, and $\lambda(D) = 0 = \lceil pW/k \rceil$. □

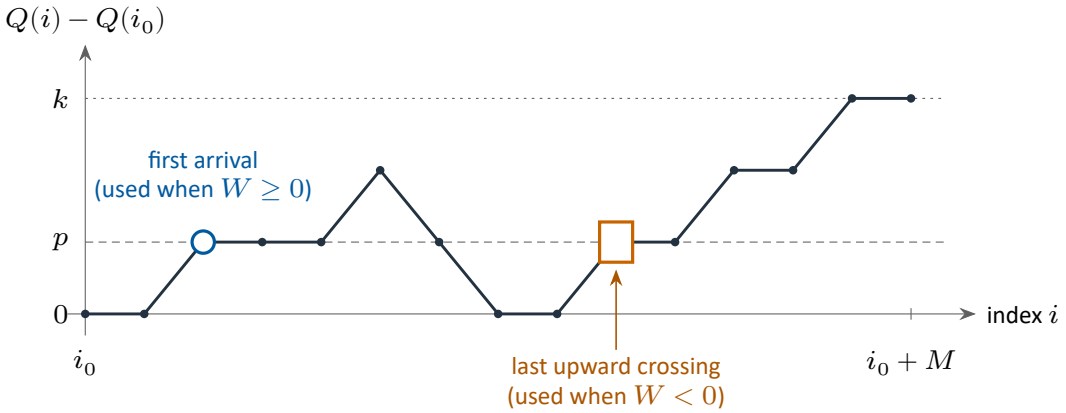


Figure 2: Where the run stops. The accumulated quota along one period of a circuit, from the starting index i_0, for a circuit with $k = 3$ and the level $p = 1$. For nonnegative winding the run stops at the first arrival at the level (circle); until then the quota is below the level. For negative winding it stops at the last upward crossing (square); from then on the quota is at or above the level. Either stop has quota exactly p and arrives at a J vertex.

Remark 4.2 (The start). The start i_0 may be a K or L vertex at a cycle position, and that freedom is needed. If the minimum of Φ is taken over the J vertices at cycle positions only, the construction fails on a circuit of a 13-vertex map, although it passed every one of the $866{,}608$ cases examined on maps with at most seven vertices (Section 10). For lower limits only the case $W \ge 1$ of the lemma is needed, since for $W \le 0$ the whole circuit serves; the case $W < 0$ is what gives upper limits and exact values. In that case the choice of the last upward crossing also matters: on a circuit of a 29-vertex map the first arrival gives too large an excess (Section 10).

5. Proofs of the theorems on endpoint totals

Theorem 5.1 (Ceiling formula). *Let I be an integer interval with $N_I \neq \emptyset$. Then*

$$\min\{\lambda(x) : x \in N_I \cap \mathbb{Z}^E\} = \lceil \min\{\lambda(x) : x \in N_I\} \rceil ,$$

where E is the set of arcs.

Proof. Let x^* be the minimiser of Lemma 3.2. An integer point of N_I has integer excess at least $\lambda(x^*)$, hence at least $\lceil \lambda(x^*) \rceil$. If x^* is an integer point it attains this bound itself. Otherwise $x^* = A + (p/k)\chi(\Gamma)$ with $k \geq 2$ and $0 < p < k$. Lemma 4.1 supplies D, and by Lemma 3.3 the integer flow $B = A + D$ has $q(B) = q(x^*) \in I$ and $\lambda(B) \leq \lceil \lambda(x^*) \rceil$. □

Proof of Theorem 1.3. Let $(g, h) \in \mathcal{K}$ have $q_S(h) \in I$. By Lemma 2.1(b) it is given by a flow with $\lambda = 0$, which lies in N_I. By Theorem 5.1 there is an integer flow in N_I with $\lambda \leq 0$. By Lemma 2.1(b) again, its pair (g, h) is an integer core certificate in the box, and its quota $q_S(h)$ lies in I. □

Proof of Corollary 1.4. $\mathcal{K}$ is a nonempty polytope, so q_S attains a minimum μ_- and a maximum μ_+ on it, and every integer point of $\mathcal{K}$ has q_S in $\mathbb{Z} \cap [\mu_-, \mu_+]$. Conversely let t be an integer in $[\mu_-, \mu_+]$. On the segment joining a minimiser to a maximiser some point of $\mathcal{K}$ has $q_S = t$, and Theorem 1.3 with $I = \{t\}$ gives an integer point of $\mathcal{K}$ with $q_S = t$. Finally Theorem 1.3 with $I = \mathbb{R}$ shows that $\mathcal{K}$ has an integer point, so $\mathbb{Z} \cap [\mu_-, \mu_+]$ is not empty and its least and greatest elements are $\lceil \mu_- \rceil$ and $\lfloor \mu_+ \rfloor$. □

Proof of Corollary 1.5. Add to the box the bounds $\lfloor g^*_v \rfloor \leq g_v \leq \lceil g^*_v \rceil$ and $\lfloor h^*_v \rfloor \leq h_v \leq \lceil h^*_v \rceil$. The result is again a box, and its core contains (g^*, h^*). Apply Theorem 1.3 to it with $I = [\lfloor q_S(h^*) \rfloor , \lceil q_S(h^*) \rceil]$. □

Proof of Theorem 1.1. Let $(g, h) \in \mathcal{R}(f, a, m, T)$. The weak components of f partition Y, and P and K_T are block diagonal with respect to them. On a component V with cycle C the restriction of (g, h) is therefore a core certificate for the restricted map and data, in the trivial box, with $q_C \geq \eta_C$. Theorem 1.3 with $S = C$ and $I = [\eta_C, \infty)$ gives an integer core certificate on V with $q_C \geq \eta_C$. The integer certificates of the components together form an integer point of $\mathcal{R}(f, a, m, T)$. □

Proof of Corollary 1.2. By Theorem 2.1 of *One Ring* a host exists if and only if the exact-cycle alternative holds or $\mathcal{R}(f, a, m, T)$ has an integer point, and by Theorem 1.1 the latter is equivalent to $\mathcal{R}(f, a, m, T) \neq \emptyset$. □

Remark 5.2 (The ceiling is needed). Let f exchange two vertices, with $T = 0$ and $a = m = 0$, and let $S = V$ and $I = [L, \infty)$ with $L \geq 0$. Then the usage at v is $g_v + h_v$, so every flow has $\lambda \geq \max(h_0, h_1) \geq (h_0 + h_1)/2$. The least real excess is $L/2$, attained at $g = 0$, $h = (L/2, L/2)$, and the least integer excess is $\lceil L/2 \rceil$. So the ceiling in Theorem 5.1 cannot be removed. Theorem 1.3 uses only that a real excess of at most zero rounds to an integer excess of at most zero.

6. Deep endpoints

Fix $\omega \in \mathbb{R}^V_{\geq 0}$ vanishing outside the deep vertices, and write $q_\omega(x) = \sum_v \omega_v\, x(h_v)$ for a flow or circulation x. An arc h_w with w deep is a *deep arc*; both its ends are at tree positions.

Lemma 6.1 (Excursions)**.** *Let Γ be a signed simple circuit with at least one vertex at a cycle position. There is an integer circulation D with properties (1) and (2) of Lemma 3.3,*

$$q_\omega(D) \geq q_\omega(\chi(\Gamma)) \qquad \textit{and} \qquad \lambda(D) \leq 0.$$

Proof. Use the notation e_i, ε_i, V_i, X of the proof of Lemma 4.1. An *excursion* is a stretch of Γ from an index $r \in X$ to the next index $r' \in X$ after it, with at least one vertex strictly between them; those vertices are at tree positions. A deep arc joins two tree positions, so each deep arc of Γ lies in exactly one excursion and is neither its first arc nor its last.

Take one excursion and let s_j be the sum of $\varepsilon_i\, \omega(e_i)$ over its first j arcs, $0 \leq j \leq r' - r$, where $\omega(e) = \omega_w$ if $e = h_w$ and $\omega(e) = 0$ otherwise. Let $\mu = \max_j s_j$; then $\mu \geq s_0 = 0$, and μ is at least the sum over the whole excursion. If $\mu > 0$, let j^* be the least j with $s_j = \mu$. The step into j^* raises the sum, so it traverses forwards a deep arc h_w with $\omega_w > 0$, and arrives at J_b with $b = \varphi(w)$ a tree position. Keep the prefix ρ formed by the first j^* arcs of the excursion. Its first arc leaves the cycle position V_r for a tree position; by (F1) it is traversed backwards and contributes at most 0 to the advance. Its other arcs join tree positions and have advance 0 by (F2). Let Π be the path of (F3) from J_b back to V_r, and let D_ρ be the sum of the signed incidence vectors of ρ and Π. It is an integer circulation with $q_\omega(D_\rho) = \mu$ and $c\,\lambda(D_\rho) = \mathsf{adv}(\rho) + \mathsf{adv}(\Pi) \leq c - 1$, so $\lambda(D_\rho) \leq 0$.

Let D be the sum of the D_ρ over the excursions with $\mu > 0$. The kept prefixes lie in different excursions, so together they use each arc of Γ at most once, with its sign in Γ; the paths Π contribute nonnegative amounts on arcs that are not g or h arcs. This gives (1) and (2). Since every deep arc of Γ lies in an excursion, $q_\omega(D)$ is the sum of the numbers μ, which is at least $q_\omega(\chi(\Gamma))$. And $\lambda(D) \leq 0$. □

Proof of Theorem 1.6. Let $N_0 = \{x \in N : \lambda(x) \leq 0\}$. By Lemma 2.1 its points give exactly the core certificates in the box, so the maximum of $\sum_v \omega_v h_v$ over $\mathcal{K}$ is the maximum of q_ω over N_0. On N_0 we have $0 \leq g_v, h_v \leq m_v$ and $0 \leq x(\psi_v) \leq m_v$, and β, σ are determined by g, h; so N_0 is a nonempty polytope and q_ω attains its maximum at a vertex x^* of N_0.

If x^* is an integer point there is nothing to prove; this includes the case that x^* is a vertex of N. Otherwise, as in the proof of Lemma 3.2 with λ in the role of q, the point x^* lies in the relative interior of an edge of N on which λ is not constant, and $\lambda(x^*) = 0$. So $x^* = A + \theta\chi(\Gamma)$ with $0 < \theta < 1$, where A and $A + \chi(\Gamma)$ are integer points of N and Γ is directed so that $W = \lambda(\chi(\Gamma)) > 0$. Then $\lambda(A) = -\theta W < 0$, so $A \in N_0$. Put $k_\omega = q_\omega(\chi(\Gamma))$.

If $k_\omega \leq 0$ then $q_\omega(A) = q_\omega(x^*) - \theta k_\omega \geq q_\omega(x^*)$, and A is an integer maximiser. If $k_\omega > 0$, note that $W \neq 0$ forces Γ to have an arc of nonzero advance, hence by (F2) a vertex at a cycle position. Lemma 6.1 gives D, and $B = A + D$ lies in N by the argument of Lemma 3.3. Now $\lambda(B) \leq \lambda(A) < 0$, so $B \in N_0$, while $q_\omega(B) \geq q_\omega(A) + k_\omega > q_\omega(A) + \theta k_\omega = q_\omega(x^*)$. This contradicts the choice of x^*, so this case does not occur. □

The difference from Lemma 4.1 is that a run which stays in a tree has no advance to pay for, so it can be closed at no cost however much quota it carries.

7. Two further totals that round

The closing lemma used two things about the selected arcs: the quota changes by one at a time, and it rises only on arrival at a vertex from which the start can be regained cheaply without using a protected arc. Two other families of arcs have these properties for lower limits.

Proposition 7.1 (Endpoints and unused capacity)**.** *Let f, T, a, m and the box be as in Theorem 1.3, let $S, R \subseteq V$, and put*

$$F(g,h) = \sum_{w \in S} h_w + \sum_{v \in R} (m_v - (g + K_T h)_v).$$

For every integer L, if some $(g,h) \in \mathcal{K}$ has $F(g,h) \geq L$ then some $(g,h) \in \mathcal{K} \cap \mathbb{Z}^{2V}$ has $F(g,h) \geq L$. Hence the largest integer value of F on $\mathcal{K}$ is the floor of the largest real value.

Proof. Let $q'(x) = \sum_{w \in S} x(h_w) + \sum_{v \in R} x(\psi_v)$. A core certificate is given by a flow with $\lambda = 0$, on which $x(\psi_v) = m_v - (g + K_T h)_v$ for every v, so $q' = F$ there. Lemma 3.2 holds for q' with the same proof. In Lemma 3.3, for the interval $[L, \infty)$, property (3) may be weakened to $q'(D) \geq p$. If $W \leq 0$, $D = \chi(\Gamma)$ has these properties. If $W \geq 1$ then X is not empty, and the first case of the proof of Lemma 4.1 applies to q': the selected arcs are h_w and ψ_v, the quota still changes by at most one per step, and it rises only on arrival at the head of a selected arc, which is a J vertex. The path Π may now contain selected arcs ψ_v, traversed forwards, so $q'(D) \geq p$; properties (1), (2) and (4) are unchanged. As in Theorem 5.1 we obtain an integer flow B with $q'(B) \geq L$ and $\lambda(B) \leq 0$. For its pair (g,h), Lemma 2.1(a) gives $m_v - (g + K_T h)_v = B(\psi_v)$ at tree vertices and $m_v - (g + K_T h)_v = B(\psi_v) - \lambda(B) \geq B(\psi_v)$ on the cycle. So $F(g,h) \geq q'(B) \geq L$. □

Proposition 7.2 (Continuing mass entering the cycle)**.** *Let f, T, a, m be as in Theorem 1.3, and let the box impose no upper bound on g_y for $y \in C$. Let S' be a set of vertices u with $f(u) \in C$. For every integer L, if some $(g,h) \in \mathcal{K}$ has $\sum_{u \in S'} g_u \geq L$ then some $(g,h) \in \mathcal{K} \cap \mathbb{Z}^{2V}$ has $\sum_{u \in S'} g_u \geq L$.*

Proof. Let $q''(x) = \sum_{u \in S'} x(g_u)$. Lemma 3.2 holds for q''. In Lemma 3.3 weaken (3) to $q''(D) \geq p$, and on the arcs g_y, $y \in C$, replace (2) by (1); this suffices there because those arcs have no upper bound. If $W \leq 0$ take $D = \chi(\Gamma)$. Let $W \geq 1$. The quota now rises only on forward traversal of a selected arc g_u, whose head $K_{f(u)}$ is a K vertex at a cycle position. Let X' be the set of indices i for which V_i is a K or L vertex at a cycle position; it is not empty because $k > 0$. Choose $i_0 \in X'$ with $\Phi(i_0)$ least and let i_1 be the least $i > i_0$ with $Q(i) = Q(i_0) + p$. Then V_{i_1} is a K vertex at a cycle position, so $i_1 \in X'$ and $\Phi(i_1) \geq \Phi(i_0)$ by the choice of i_0, which gives (9). Close the run from V_{i_1} to V_{i_0} by the forward path that alternates the arcs $\beta_v : K_v \to L_v$ and $g_v : L_v \to K_{f(v)}$ with $v \in C$, ending with an arc β_v if V_{i_0} is an L vertex. It has advance at most $c - 1$, contains no h arc and no arc g_u with u a tree vertex, and any selected arc on it adds to the quota. The rest is as in Lemma 4.1 and Theorem 5.1. □

Remark 7.3 (A common head or a common tail). If the selected arcs, of either kind, all have the same head, or all have the same tail, insert one new vertex and one new arc through which they all pass. Their sum is then the flow on a single arc, and an interval on the sum is a pair of bounds on that arc. The result is again a flow polyhedron with integer data, so the least excess is attained at an integer flow. Such a group therefore rounds for lower limits, upper limits and exact values with no ceiling at all, and several disjoint groups of this kind can be constrained together. A single variable is the simplest case; it is Theorem 4.2 of *One Ring*.

8. Where the statements stop

8.1 Exact examples

Each row of Table 1 is an instance with a real core certificate that satisfies the stated condition and no integer core certificate that does. Maps are written as the list $(f(0), f(1), \dots)$. In every example of this section the demands on the cycles are zero, so the quota (4) imposes nothing and the core certificates are the certificates of *One Ring*. Every example was checked in exact arithmetic: the real certificate against each inequality, and the integer side by listing all integer core certificates.

What fails	Instance	Real certificate, and the integer side
A total across two components	$f = (0, 1), T = 1, a = 0,$ $m = (1, 1)$	$g = 0, h = (\frac{1}{2}, \frac{1}{2})$ has $h_0 + h_1 = 1$. Capacity is $g_v + 2h_v \le 1$, so every integer certificate has $h = 0$.
A weighted total	$f = (0), T = 1, a = 0, m = 1$	$g_0 = 0, h_0 = \frac{1}{2}$ has $2h_0 = 1$. In integers $h_0 = 0$.
A total mixing the two lists	$f = (0, 0), T = 1, a = 0,$ $m = (1, 1)$	$g = (0, \frac{1}{2}), h = (\frac{1}{2}, 0)$ has $g_1 + h_0 = 1$. In integers $g_1 + h_0 = 0$.
Continuing variables along a tree path	$f = (0, 0, 1), T = 3, a = 0,$ $m = (2, 1, 1)$	$g = (0, \frac{1}{2}, \frac{1}{2}), h = (\frac{1}{2}, 0, 0)$ has $g_1 + g_2 = 1$. In integers $g_1 + g_2 = 0$. Here $f(2) \notin C$.
Continuing variables with one upper bound	$f = (1, 2, 0), T = 2, a = 0,$ $m = (1, 1, 1)$, bound $g_0 \le 0$	$g = (0, \frac{1}{2}, \frac{1}{2}), h = (\frac{1}{2}, 0, 0)$ has $g_1 + g_2 = 1$. In integers $g_1 + g_2 = 0$.
Two lower limits on two endpoint sets	$f = (1, 2, 3, 0), T = 1, a = 0,$ $m = (1, 1, 1, 1)$	$g = 0, h = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ has $h_0 + h_2 = 1$ and $h_1 + h_3 = 1$. No integer certificate has both sums at least 1.
A lower and an upper limit on two endpoint sets, with two variables fixed	$f = (1, 2, 3, 0, 1, 4, 3, 6),$ $T = 1, m = \mathbf{1}, a_5 = a_7 = 1$ and $a_v = 0$ otherwise, bounds $g_1 \le 0, g_3 \le 0$	$g = (0, 0, 0, 0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}),$ $h = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, 0, \frac{1}{2})$ has $h_0 + h_2 = 1$ and $h_4 + h_5 + h_6 + h_7 = 1$. No integer certificate has $h_0 + h_2 \ge 1$ and $h_4 + h_5 + h_6 + h_7 \le 1$ together.

Table 1: Statements that fail. The first row shows that connectedness is needed for a single total in Theorem 1.3; the second that arbitrary weights cannot replace the unit coefficients of the selected endpoint variables; the third to fifth that the hypotheses under which continuing variables are allowed in Proposition 7.2 cannot simply be dropped; the last two that one interval on one endpoint set cannot in general be replaced by two conditions on two sets.

In the last example, dropping either of the two bounds, or either of the two limits, leaves an instance that has an integer certificate. The same failure occurs with the upper limit on deep endpoints only. Take $f = (1, 2, 3, 0, 1, 4, 5, 3, 7, 8), T = 1, m = \mathbf{1}, a_6 = a_9 = 1$ and $a_v = 0$ otherwise, with the bounds $g_1, g_3 \le 0$ and $h_4, h_5, h_7, h_8 \le 0$, and ask for $h_0 + h_2 \ge 1$ and $h_6 + h_9 \le 1$. The point with $g_v = \frac{1}{2}$ for $v \ge 4$, $h_v = \frac{1}{2}$ for $v \in \{0, 1, 2, 3, 6, 9\}$ and all other entries 0 is a real core certificate satisfying both conditions, and none of the integer core certificates does.

Two further examples mark the boundary of the positive results themselves.

Minima of deep endpoints are not integers. Let $f = (0, 0, 1, 2), T = 1, a = (0, 0, 0, 1),$ $m = \mathbf{1}$, with the bounds $h_1 \le 0$ and $h_2 \le 0$. Vertex 3 is deep. Capacity at the fixed point gives $2h_0 \le 1$; matching gives $g_1 \le h_0$, and then, since $h_1 = h_2 = 0$, $g_2 \le g_1$ and $g_3 \le g_2$;

demand gives $h_3 \geq 1 - g_3 \geq \frac{1}{2}$. The value $\frac{1}{2}$ is attained by $g = (0, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, $h = (\frac{1}{2}, 0, 0, \frac{1}{2})$. In integers $h_0 = 0$, so $h_3 \geq 1$, and $h_3 = 1$ is attained. Thus Theorem 1.6 does not extend to minima, or to weights of both signs, and the ceiling in Corollary 1.4 is needed even for a single deep endpoint.

Fractional vertices remain. The example of Section 6.2 of *One Ring*, used in the Overview, has a fractional vertex at which $\sum_v h_v = 11/2$. So $\mathcal{K}$ is not an integral polytope, and an arbitrary linear objective need not have an integer optimum.

8.2 Open questions

Both questions concern two conditions at once in one connected component, with no extra bounds on individual variables.

1. *Cycle endpoints and entering mass.* Let E be the set of tree vertices u with $f(u) \in C$. If a real core certificate has $\sum_{y \in C} h_y \geq L_1$ and $\sum_{u \in E} g_u \geq L_2$ for integers $L_1 > L_2$, must an integer one? The case $L_1 \leq L_2$ follows from Proposition 7.2, because matching summed around the cycle gives $\sum_{u \in E} g_u \leq \sum_{y \in C} h_y$.
2. *A lower and an upper limit.* Let $S_1, S_2 \subseteq V$ be disjoint. If a real core certificate has $q_{S_1} \geq L$ and $q_{S_2} \leq U$, must an integer one? This is true when all demands are at most zero: set $g = 0$ and $h_w = 0$ for $w \notin S_1$, which keeps the first condition and a core certificate, and apply Theorem 1.3 with those variables fixed at zero; then $q_{S_2} = 0 \leq U$. With positive demands it is open, and the last row of Table 1 shows that it fails once two variables are fixed.

Random searches for a counterexample to either question, described in Section 10, found none.

9. Relationship to established results

Special cases reachable from circular-ones theory. On a pure cycle the matrix K_T is a multiple of the all-ones matrix plus a matrix whose rows are circular intervals. Bartholdi, Orlin and Ratliff [2] solve covering programs with circular ones by fixing the total of the variables, which leaves a network problem, and Hochbaum and Levin [7] extend the method to mixed covering and packing constraints. With partial sums as variables and the total held at an integer, every constraint becomes a difference of two variables with an integer bound (Section 2 of [7]). Two special cases of Theorem 1.3 on a pure cycle can be reached in this way. With demands at most zero and no bounds on individual variables, one may set $g = 0$ and drop the unselected endpoint variables, and a lower limit on any set of endpoints follows. With arbitrary integer demands, eliminating the continuing variables by hand leaves circular-interval conditions on the endpoint variables, and a prescribed integer value of the total of all the endpoints follows. We do not reproduce these reductions. These sources concern variables in one cyclic order. They do not treat the trees attached to the cycle, and the general statements of Section 1 are not obtained from them by any reduction known to us.

Standard ingredients. Integrality of flow polyhedra and the description of their edges by circuits are classical. That one extra linear condition leaves a fractional optimum on a single edge between integer flows is used, for example, by Panda et al. [9] (their Lemma 4.4) and, for general totally unimodular systems, by Aprile et al. [1]. Arguments that organise residual circulations by a winding number appear in Colin de Verdière and Schrijver [6], for planar graphs

and unit windings. None of these steps is claimed as new here. What is specific to the present network is Lemma 4.1: a run of the fractional circuit with exactly the missing quota can be closed, through arcs with no upper bounds, at a cost that the potential Φ controls.

Why general frameworks do not give the result directly. A flow problem with one extra unweighted condition does not in general have the ceiling property of Theorem 5.1. On two opposite arcs a circulation is $x_1 = x_2 = t \geq 0$; under $x_1 + x_2 \geq 1$ the least value of $x_1 + x_2$ is 1 over the reals and 2 over the integers. The certificate polytopes are not integral: for one fixed point with $T = 1$ and $m = 3$, the conditions $g \geq 0$, $h \geq 1$, $g + 2h \leq 3$ have the vertex $(0, \frac{3}{2})$. So this system, with its integer right-hand sides, is not totally dual integral, and not box-totally dual integral in the sense of [5]. The least excess as a function of the quota is $\lceil L/2 \rceil$ in Remark 5.2, with values $0, 1, 1$ at $L = 0, 1, 2$, which is not discretely convex in the sense surveyed by Murota [8].

Extent of the literature check. The comparisons above come from a check of twenty-six sources, twenty-two of them read in full, carried out during this work, together with a separate reading of [7]. Within that set we found no result implying the case with trees.

10. What has been checked, and how

The general statements of this paper rest on the written proofs. This section records which proofs have been read independently, what has been formalised, and which computations were run. The computations test the constructions and the statements on finite families; the proofs do not depend on them. The archive deposited on Zenodo with this paper under DOI 10.5281/zenodo.24121981 contains the LaTeX source, the Lean files described below, and the programs and recorded outputs behind Table 2, except its eighth row, whose earlier runs are retained in the author's research record.

10.1 Independent readings

- *The aggregate implication* (Theorem 1.1) was first proved by a direct argument for a unit quota. That proof was read by two independent AI referees, each of which also implemented the construction and ran it in exact arithmetic, and in a third review by a separate AI system.
- *Lower limits for every endpoint set* (Theorem 1.3 for intervals $[L, \infty)$, with Lemmas 3.2, 3.3 and the case $W \geq 1$ of Lemma 4.1) was read line by line in a separate AI review that was asked to break it. Its verdict was that the proof is correct and needs no repair.
- *Upper limits and exact values* (the case $W < 0$ of Lemma 4.1), *Theorem 1.6* and *Proposition 7.1* were found in that review. Each was then read line by line by a different AI system, and its construction was rebuilt and tested with separate code, as listed below.
- *Proposition 7.2* is an earlier result of this work. In a longer form, and without the box, it had independent readings together with the first proofs of Theorem 1.1. The short proof given here is new to this paper. Its construction has been machine-tested, and the proof was read in the review described in the next item, which found it valid.
- *The manuscript as a whole* was read in a further AI review, which also checked every displayed example in exact arithmetic with its own code. It reported no substantive defect in the proofs and asked for corrections of scope and wording, which have been made.

10.2 Formal verification

The aggregate implication of Theorem 1.1, for demands and inventories in the natural numbers, which include all the data of *One Ring*, is formalised in Lean 4 (version 4.19.0) as the statement

`OneRing.GeneralAggregate.real_iff_numeric`,

together with the corollary that a real certificate and a positive total inventory give a host. The statement reads: for every finite type V, every map $f : V \to V$, every horizon T and all $a, m : V \to \mathbb{N}$, real vectors g, h forming a `RealNumericCertificate` exist if and only if natural-number vectors g, h forming a `NumericCertificate` exist. These two predicates are that development's formal versions of the system (1)–(4) over the reals and over the natural numbers; their agreement with the displayed system was reviewed when the development was made. The kernel reports only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The formal proof follows a different argument from Section 4: it normalises the selected endpoint mass to one and reroutes it through a common gate. It extends a Lean development of results of *One Ring* that is larger than the three Lean files in the public archive of that paper; that development was rebuilt from source before the extension was made. The archive of the present paper holds a Lean project with the new files and every file of that development on which the statement depends, 38 files in all, with the Lean version and the mathlib commit fixed, so that the statement can be rebuilt.

Theorems 1.3, 5.1 and 1.6, the corollaries on ranges and boxes, and the two propositions are not formalised.

10.3 Computations

All arithmetic in the checks below is exact. A floating-point solver was used only to propose points, each of which was then verified in integers or rationals.

The exhaustive family. It consists of every connected map on at most seven vertices in a standard form: the cycle is $0 \to 1 \to \ldots \to c-1 \to 0$, and every tree vertex maps to a vertex with a smaller label. Maps are not reduced further by isomorphism, so some shapes occur more than once. For each map and horizon the circuits are all simple circuits of the network regarded as an undirected multigraph, loops and pairs of parallel or opposite arcs included, each taken in both directions. For each circuit, every subset of the endpoint arcs lying on it is taken as the selected set whenever that gives $k \geq 2$, with every p and every start the proof permits: every minimiser of Φ, or every vertex at the shallowest position when the circuit meets no cycle position.

What the data comparisons certify. In the rows that compare real and integer certificates on random data, the real certificate is a rational point that a floating-point solver proposed as an optimum and that was then verified exactly as a certificate. Its optimality was not certified, and is not needed: each comparison is an exact instance of the implication from that real certificate to an integer one. In the examples of Section 8 the integer side is settled by complete enumeration, and the one real minimum that is claimed is proved in the text and by exact dual multipliers.

Why larger maps were tested. A version of Lemma 4.1 that takes the start among the J vertices only passed all $866{,}608$ cases examined on maps with at most seven vertices, and a first random sample on larger maps. A counting argument showed where it could fail, and hand-built circuits on maps with 14 and 13 vertices confirm that it does: the run from the best J start has excess 2 where 1 is allowed, and the start of Lemma 4.1 is a K vertex. Agreement

What was built or compared	Scope	Result
The circulation D of Lemma 4.1, as an explicit integer vector from every permitted start; balance at every vertex and properties (1) to (4) checked	Every map with at most 7 vertices, horizons 0 to 4, every circuit in both directions, every choice of selected endpoint arcs on it with $k \geq 2$: $19{,}268{,}826$ constructions, of which $784{,}114$ have $W < 0$ and $1{,}367$ lie on circuits that meet no cycle position	All pass
The same on sampled circuits	$16{,}902$ random maps with 12 to 30 vertices, two runs: $3{,}085{,}658$ constructions, $419{,}436$ with $W < 0$, windings from -16 to 16, circuits of up to 78 arcs	All pass
The circulation of Lemma 6.1	The same maps and circuits, weights 0 or 1 on every set of deep arcs of the circuit and random weights up to 4: $2{,}371{,}880$ cases	All pass
The circulation in the proof of Proposition 7.2	The same exhaustive family: $19{,}236{,}872$ constructions; sampled: $464{,}202$	All pass
Theorem 1.3 and Corollaries 1.4, 1.5 on data: each real certificate an exact rational point, each integer certificate checked exactly	$20{,}681$ random feasible instances with 4 to 11 vertices, $15{,}098$ with random integer bounds: $92{,}646$ upper limits, $92{,}646$ lower limits, $268{,}848$ exact values, $93{,}159$ roundings inside a box	No failure
Theorem 1.6 and Proposition 7.1 on the same instances	$9{,}403$ weighted deep maxima; $20{,}681$ totals with unused capacity	No failure
The same comparison for two statements known to be false	Totals of continuing variables over the trees; endpoint totals with weights 1 and 2	233 and 625 failures found
Lower limits, before the proof was found	$384{,}204$ comparisons on $86{,}062$ random instances; all $42{,}061$ endpoint sets on maps with at most 7 vertices and horizons 1 to 3 that an earlier criterion left undecided, by integer programming with exact verification	No failure
The examples of Section 8	Rational certificates and, where needed, dual multipliers in exact fractions; every integer core certificate listed	As stated

Table 2: Computations. Counts are numbers of constructions or comparisons, not of independent instances; circuits of small maps overlap heavily. The sampled circuits were produced by a randomised depth-first search and are not a uniform sample.

on small cases does not by itself establish a construction, which is why the tests were extended to larger maps and the proofs were sent for a hostile reading.

The last upward crossing is needed. For $W < 0$ the first arrival passed every case of the exhaustive family and of the first sampled run. The second sampled run found a circuit on a 29-vertex map with $W = -2$, $k = 3$ and $p = 2$ on which the first arrival exceeds the allowed excess -1, while the last upward crossing meets it.

Every solver output was verified. With presolve enabled, the integer-programming routine returned values one below the true integer maximum on a few instances. Every integer point counted above was verified exactly, and apparent gaps were re-examined by complete enumeration.

The open questions. One of the reviews searched the two open questions of Section 8 on $6{,}049$ and $8{,}442$ random instances, with exact certificates, and found no counterexample.

Acknowledgements and AI-assistance statement

This paper arose from a programme conceived by the author, based on his independently developed *Something is happening: Occurrence Theory* [4], where he identified the problem of reconciling finite recurrent dynamics with apparently irreversible higher-level processes. He directed AI systems to seek interesting, potentially original mathematics addressing it, and set the scope and stopping criteria. AI agents then defined the specific fixed-capacity circular-realisation problem, formulated and proved its theorems, wrote the software, ran the analyses and drafted the initial manuscript. The primary models were Claude's Fable 5.2 and 5.1 and GPT-6 Astra Pro with contributions from Claude's Opus 5.5 and GPT-6.1sol models. The author reviewed stage summaries, approved or redirected investigations, commissioned critical reviews, and edited and approved the final manuscript. He deposited the release and accepts responsibility for its claims, interpretations and remaining errors. The author retains supporting project and review records; the release archive documents reviewed versions, checks and their scope.

11. Conclusion

In plain terms: *One Ring* reduced the question of whether one cycle can host a visible process to a list of inequalities in whole numbers, and left open whether solving them with fractions is already enough. It is. For every finite visible rule, every duration and all whole-number data, a real solution of the four families of inequalities gives a whole-number solution (Theorem 1.1).

The reason is a rounding property of the first three families that goes well beyond the question asked. Within one connected part of the visible rule, with those three families in force, any one total of endpoint counts can be held to a lower limit, an upper limit or an exact value in whole numbers whenever it can be in real numbers, and the whole-number totals fill the real range with no gaps (Theorem 1.3, Corollary 1.4). Endpoints that lie far out on a branch reach their largest real total exactly (Theorem 1.6). The proof is constructive: it walks part of the way round one closed walk in a network and returns along spare capacity, choosing its start by a potential and its stop by a first or a last crossing.

The statements have sharp edges. Totals across separate components, weighted totals, totals of continuing counts along a branch and two limits on two totals can fail, each by an explicit small example (Section 8). Two questions about two simultaneous conditions remain open. The aggregate implication is also proved in Lean 4, and Section 10 describes the independent readings and computations behind every result.

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